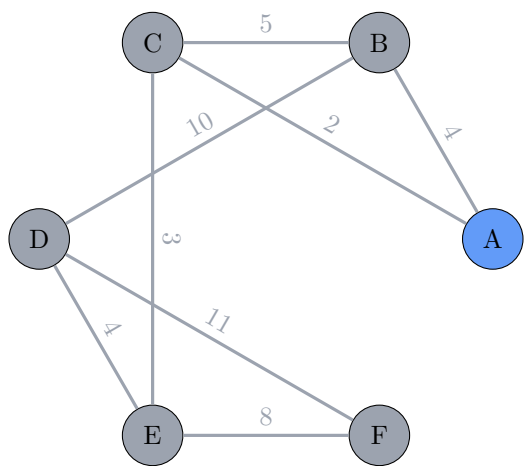


Dijkstra

Log

Initial state

Graph



Open Set



Closed Set

Node Costs

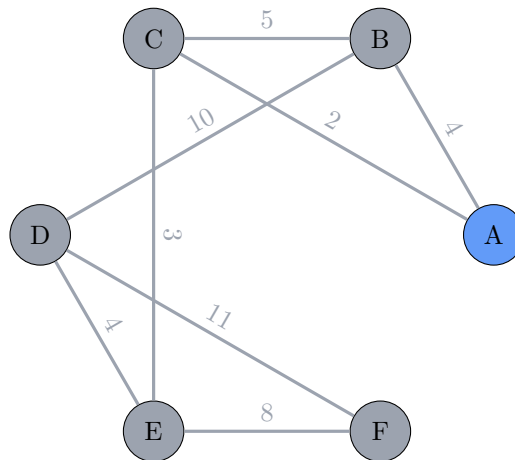
0	0	0	0	0	0
A	B	C	D	E	F

Dijkstra

Log

Sorting the open set so the lowest-cost node comes first.

Graph



Open Set



Closed Set

Node Costs

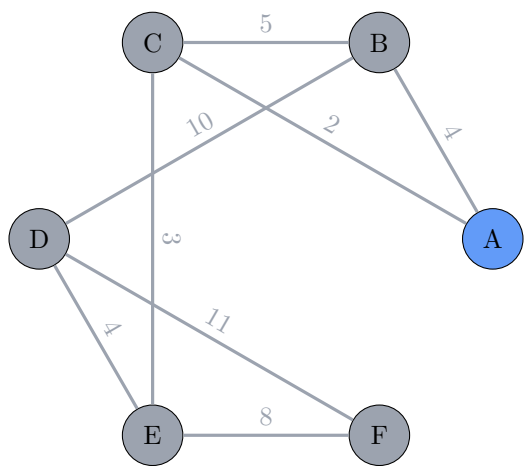
0	0	0	0	0	0
A	B	C	D	E	F

Dijkstra

Log

Grabbing the lowest-cost node from the open set.

Graph



Open Set



Closed Set

Node Costs

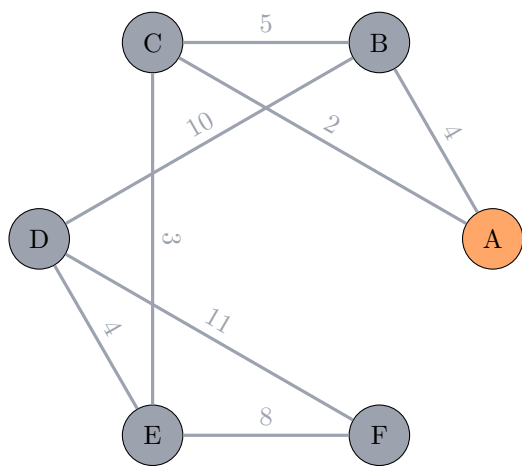
0	0	0	0	0	0
A	B	C	D	E	F

Dijkstra

Log

Visiting node A and moving it to the closed set.

Graph



Open Set

Closed Set



Node Costs

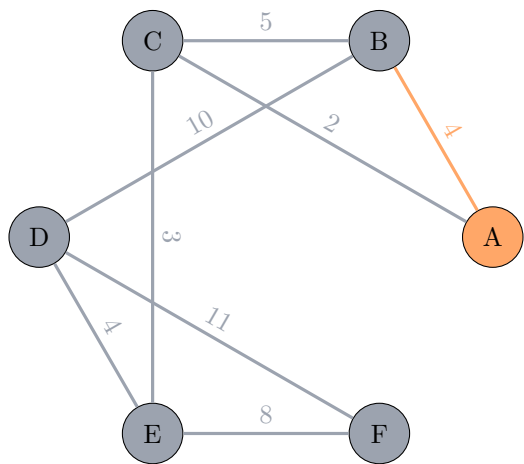
0	0	0	0	0	0
A	B	C	D	E	F

Dijkstra

Log

Looking at neighbor B | is going through A cheaper than what we already know?

Graph



Open Set

Closed Set



Node Costs

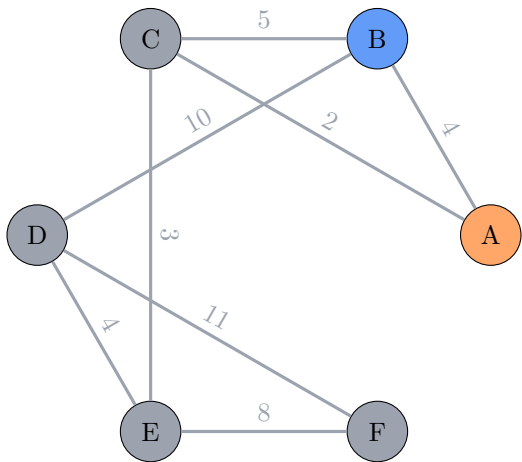
0	0	0	0	0	0
A	B	C	D	E	F

Dijkstra

Log

Neighbor B hasn't been seen yet | adding it to the open set.

Graph



Open Set



Closed Set



Node Costs

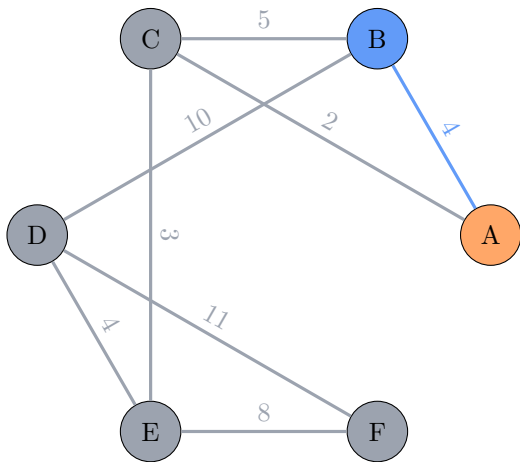
0	0	0	0	0	0
A	B	C	D	E	F

Dijkstra

Log

Found a cheaper path to B through A | updating its cost to 4.

Graph



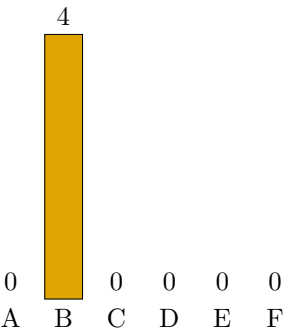
Open Set



Closed Set



Node Costs

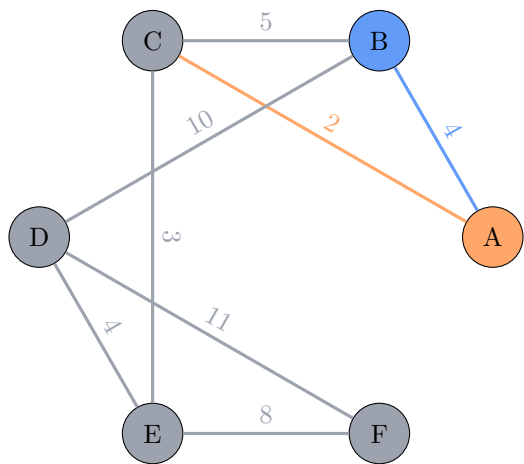


Dijkstra

Log

Looking at neighbor C | is going through A cheaper than what we already know?

Graph



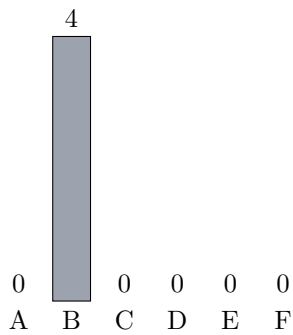
Open Set



Closed Set



Node Costs

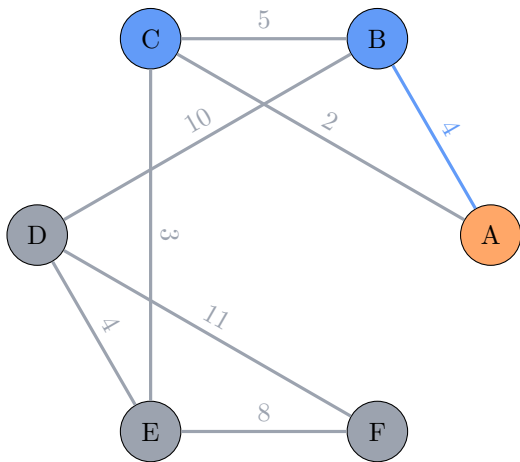


Dijkstra

Log

Neighbor C hasn't been seen yet | adding it to the open set.

Graph



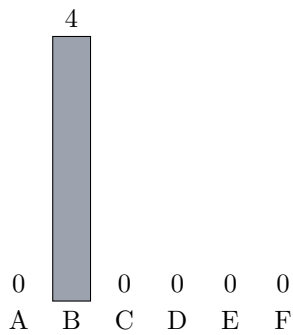
Open Set



Closed Set



Node Costs

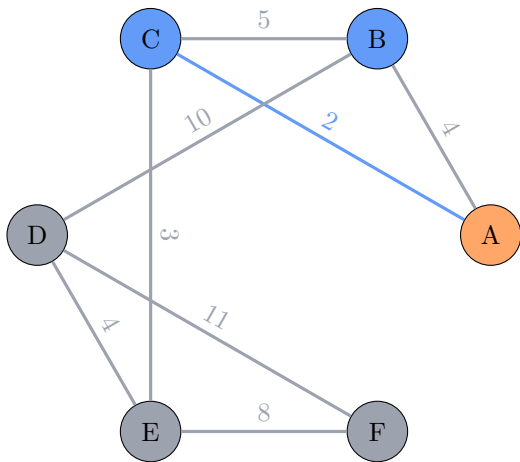


Dijkstra

Log

Found a cheaper path to C through A | updating its cost to 2.

Graph



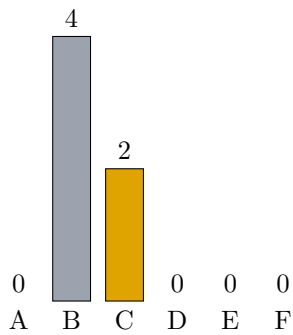
Open Set

B	C
---	---

Closed Set

A

Node Costs

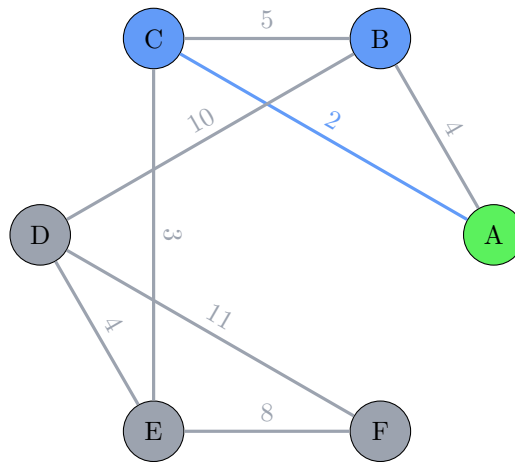


Dijkstra

Log

Sorting the open set so the lowest-cost node comes first.

Graph



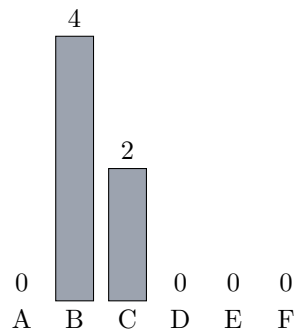
Open Set



Closed Set



Node Costs

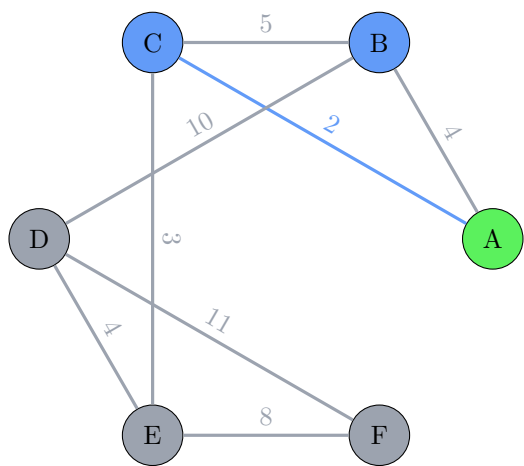


Dijkstra

Log

Grabbing the lowest-cost node from the open set.

Graph



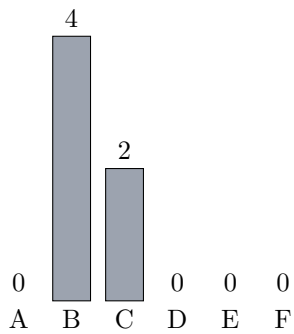
Open Set



Closed Set



Node Costs

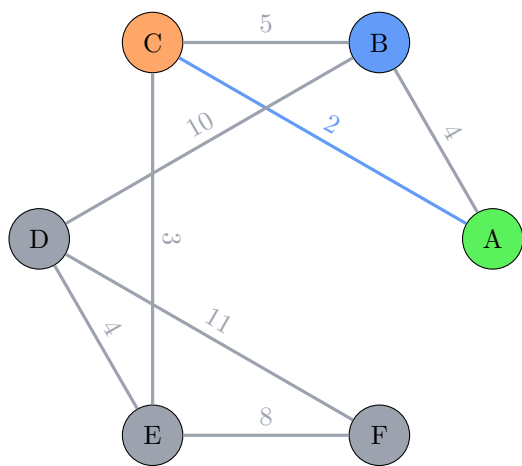


Dijkstra

Log

Visiting node C and moving it to the closed set.

Graph



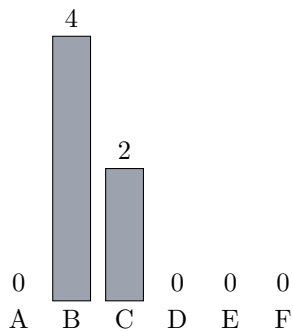
Open Set



Closed Set



Node Costs

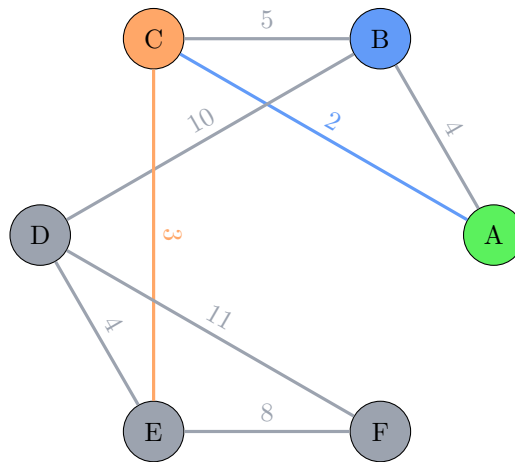


Dijkstra

Log

Looking at neighbor E | is going through C cheaper than what we already know?

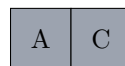
Graph



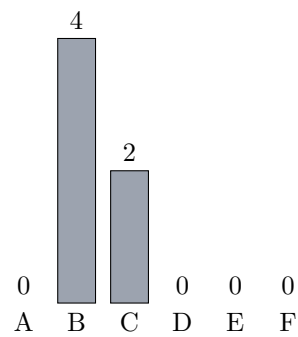
Open Set



Closed Set



Node Costs

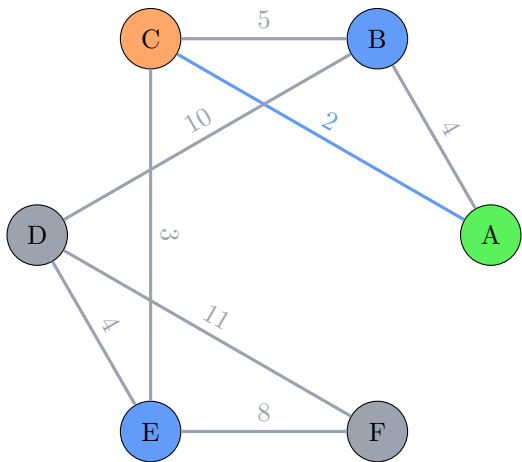


Dijkstra

Log

Neighbor E hasn't been seen yet | adding it to the open set.

Graph



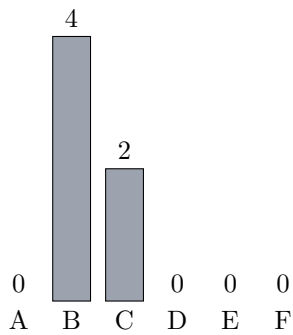
Open Set



Closed Set



Node Costs

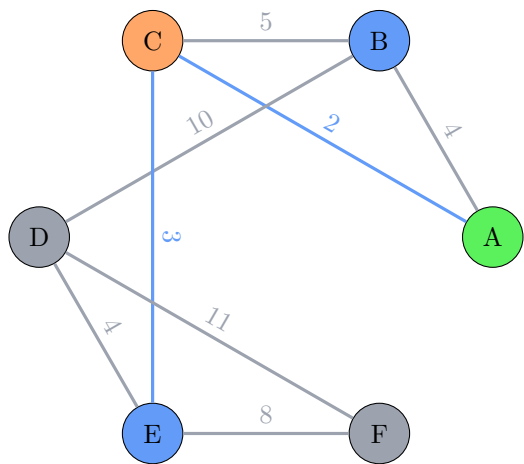


Dijkstra

Log

Found a cheaper path to E through C | updating its cost to 5.

Graph



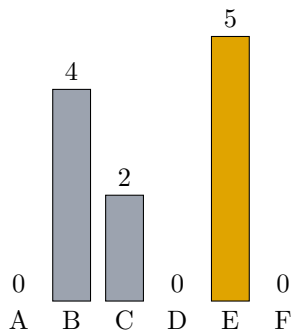
Open Set

B	E
---	---

Closed Set

A	C
---	---

Node Costs

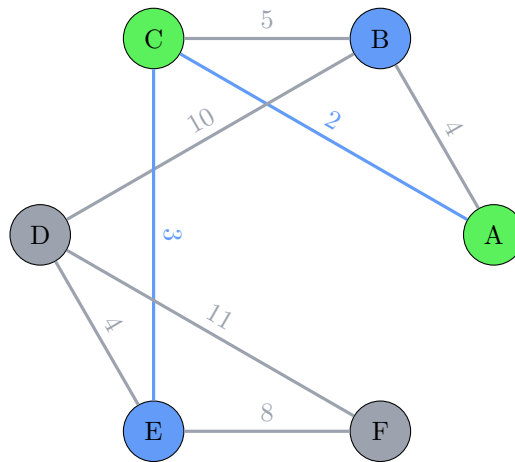


Dijkstra

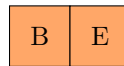
Log

Sorting the open set so the lowest-cost node comes first.

Graph



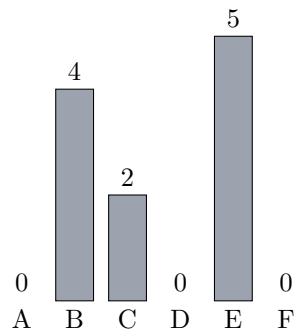
Open Set



Closed Set



Node Costs

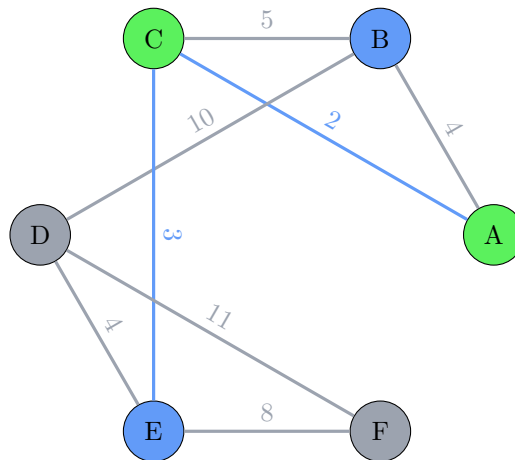


Dijkstra

Log

Grabbing the lowest-cost node from the open set.

Graph



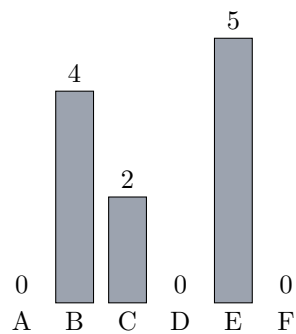
Open Set



Closed Set



Node Costs

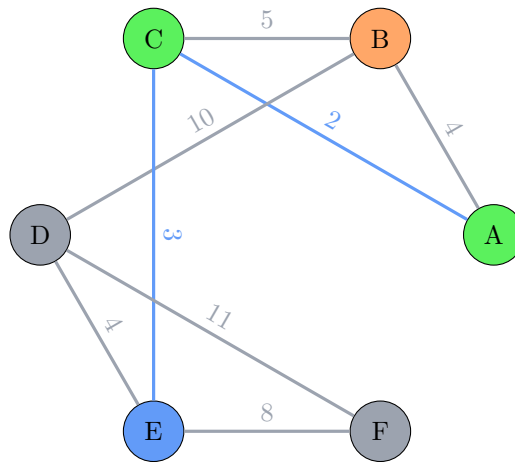


Dijkstra

Log

Visiting node B and moving it to the closed set.

Graph



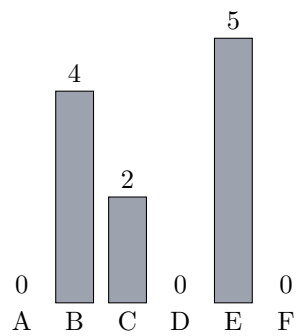
Open Set



Closed Set



Node Costs

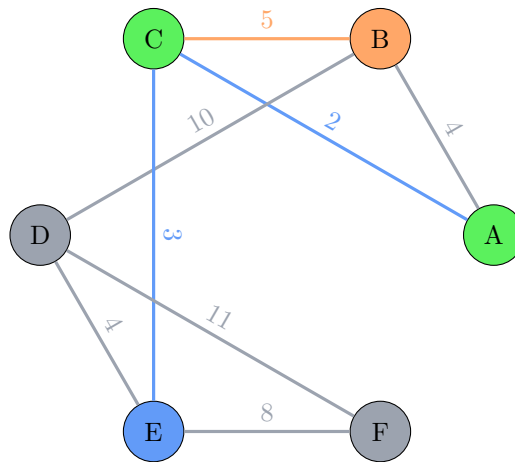


Dijkstra

Log

Looking at neighbor C | is going through B cheaper than what we already know?

Graph



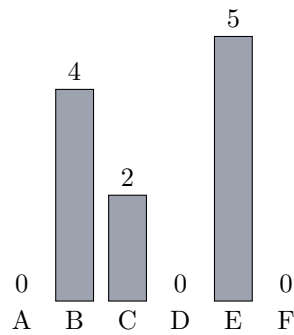
Open Set

E

Closed Set

A C B

Node Costs

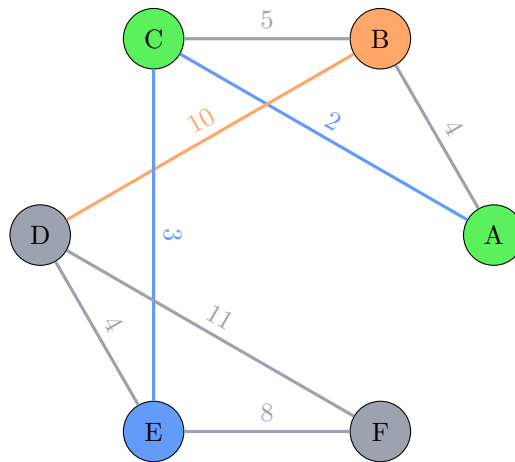


Dijkstra

Log

Looking at neighbor D | is going through B cheaper than what we already know?

Graph



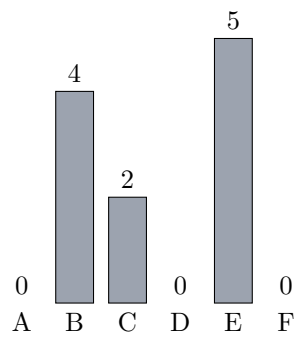
Open Set

E

Closed Set

A	C	B
---	---	---

Node Costs

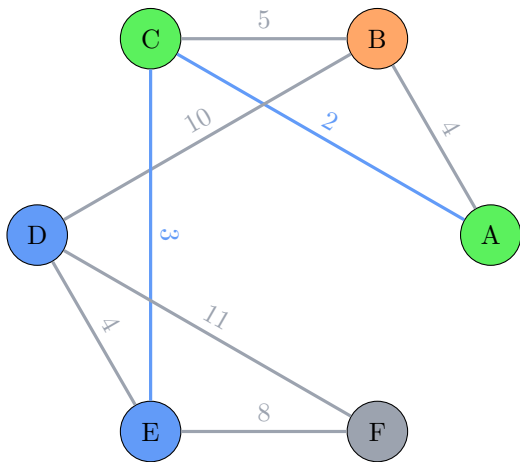


Dijkstra

Log

Neighbor D hasn't been seen yet | adding it to the open set.

Graph



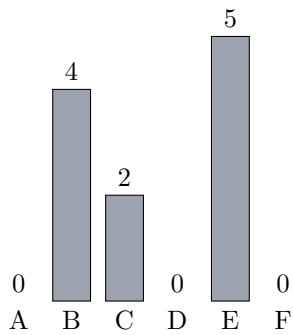
Open Set



Closed Set



Node Costs

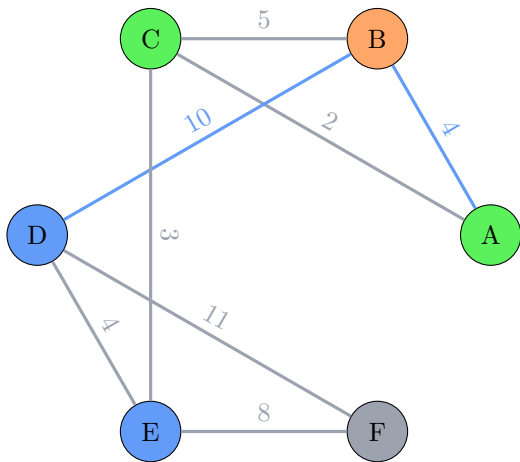


Dijkstra

Log

Found a cheaper path to D through B | updating its cost to 14.

Graph



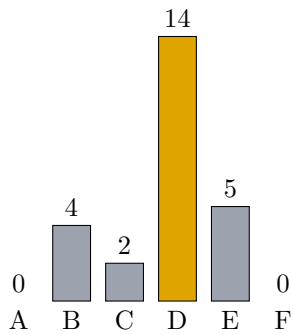
Open Set

E	D
---	---

Closed Set

A	C	B
---	---	---

Node Costs

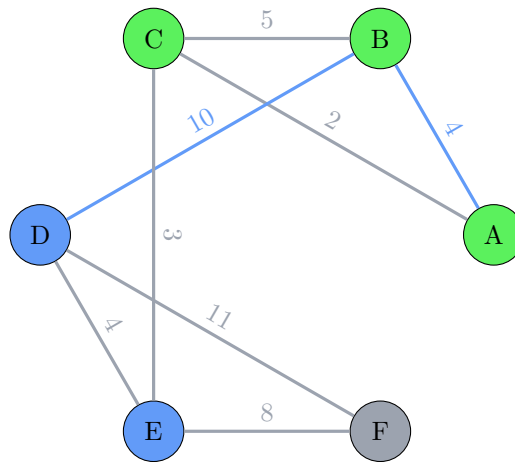


Dijkstra

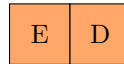
Log

Sorting the open set so the lowest-cost node comes first.

Graph



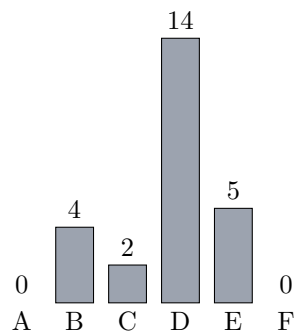
Open Set



Closed Set



Node Costs

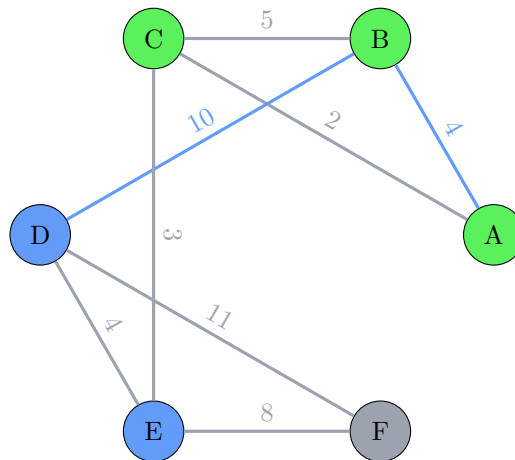


Dijkstra

Log

Grabbing the lowest-cost node from the open set.

Graph



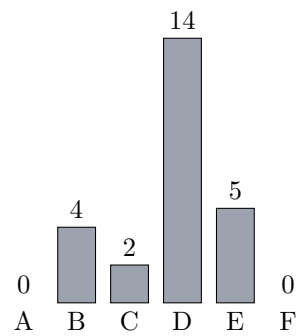
Open Set



Closed Set



Node Costs

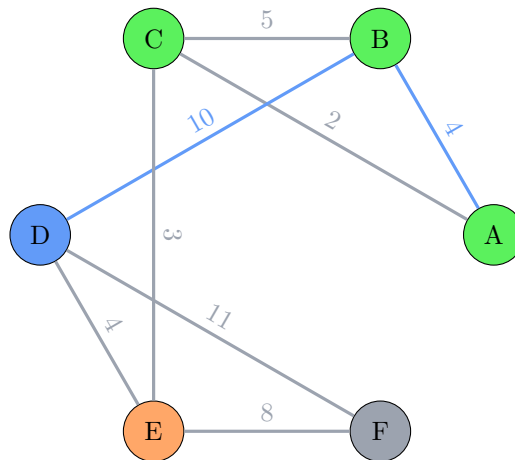


Dijkstra

Log

Visiting node E and moving it to the closed set.

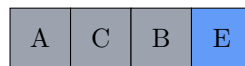
Graph



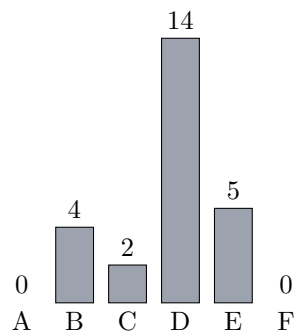
Open Set



Closed Set



Node Costs

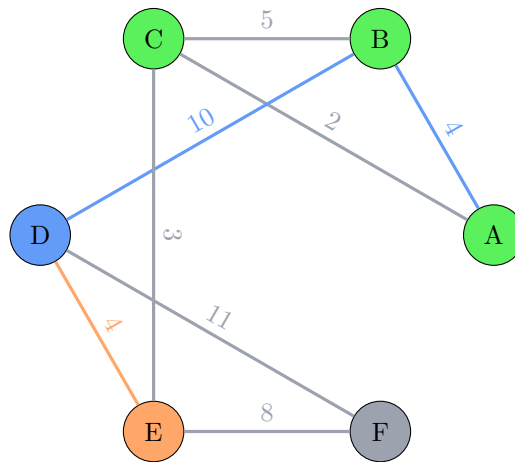


Dijkstra

Log

Looking at neighbor D | is going through E cheaper than what we already know?

Graph



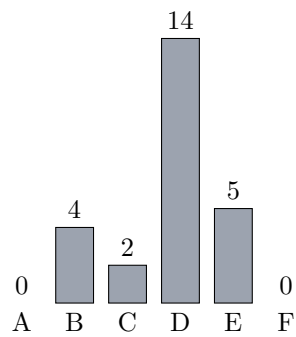
Open Set

D

Closed Set

A	C	B	E
---	---	---	---

Node Costs

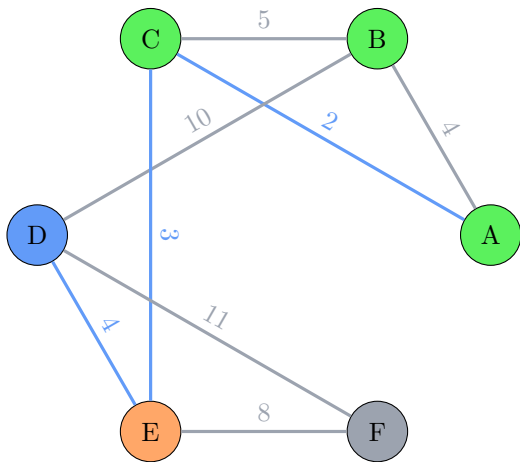


Dijkstra

Log

Found a cheaper path to D through E | updating its cost to 9.

Graph



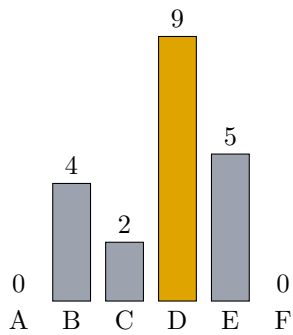
Open Set



Closed Set



Node Costs

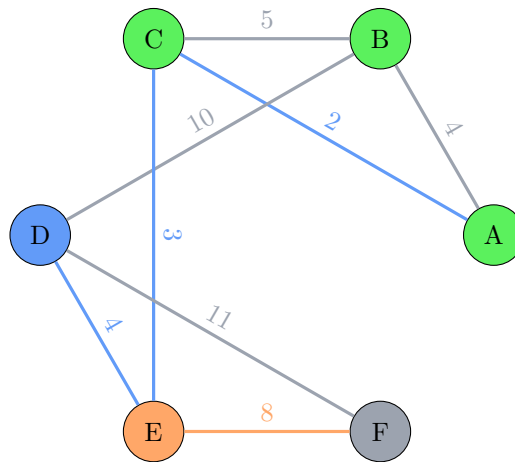


Dijkstra

Log

Looking at neighbor F | is going through E cheaper than what we already know?

Graph



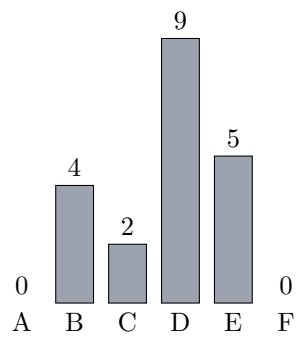
Open Set

D

Closed Set

A	C	B	E
---	---	---	---

Node Costs

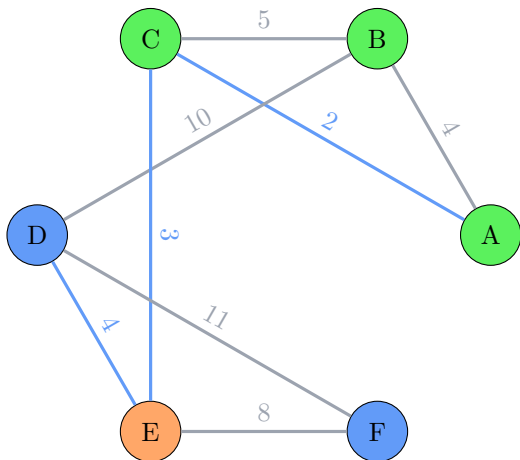


Dijkstra

Log

Neighbor F hasn't been seen yet | adding it to the open set.

Graph



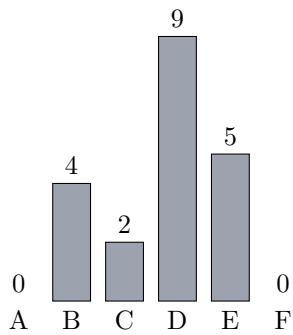
Open Set



Closed Set



Node Costs

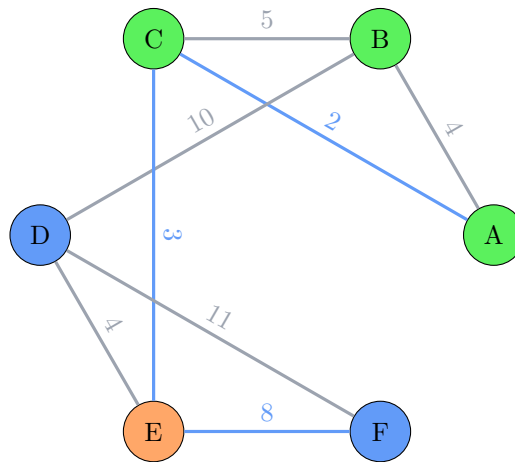


Dijkstra

Log

Found a cheaper path to F through E | updating its cost to 13.

Graph



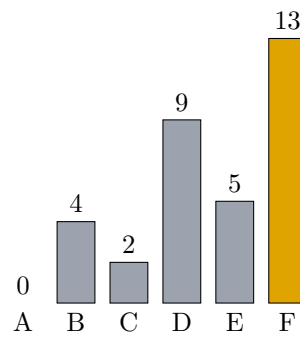
Open Set

D	F
---	---

Closed Set

A	C	B	E
---	---	---	---

Node Costs

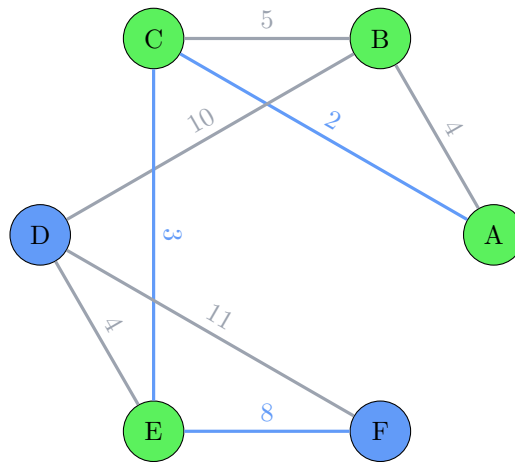


Dijkstra

Log

Sorting the open set so the lowest-cost node comes first.

Graph



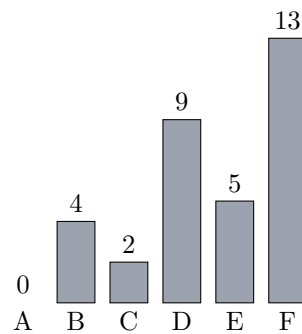
Open Set

D	F
---	---

Closed Set

A	C	B	E
---	---	---	---

Node Costs

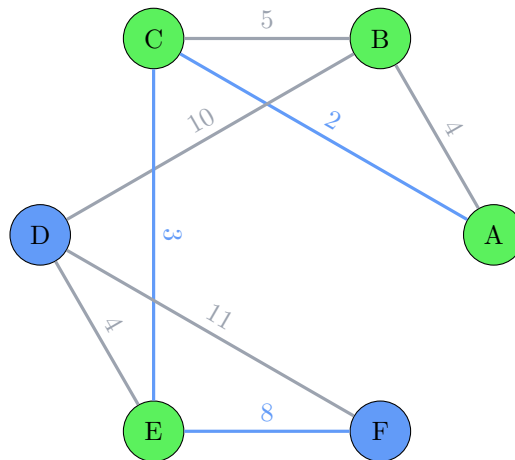


Dijkstra

Log

Grabbing the lowest-cost node from the open set.

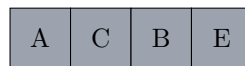
Graph



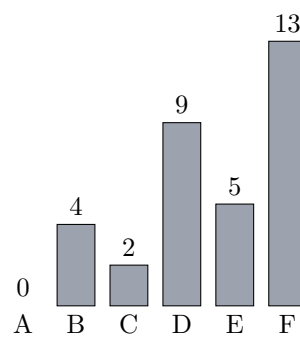
Open Set



Closed Set



Node Costs

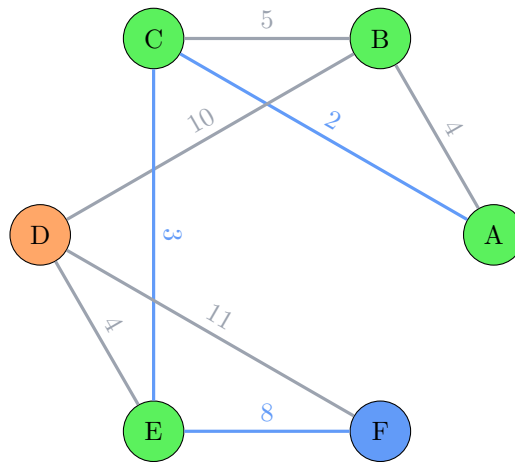


Dijkstra

Log

Visiting node D and moving it to the closed set.

Graph



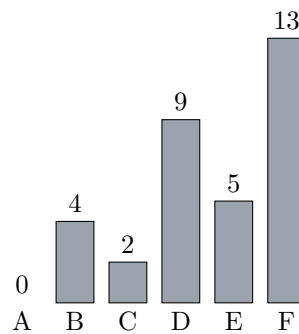
Open Set



Closed Set



Node Costs

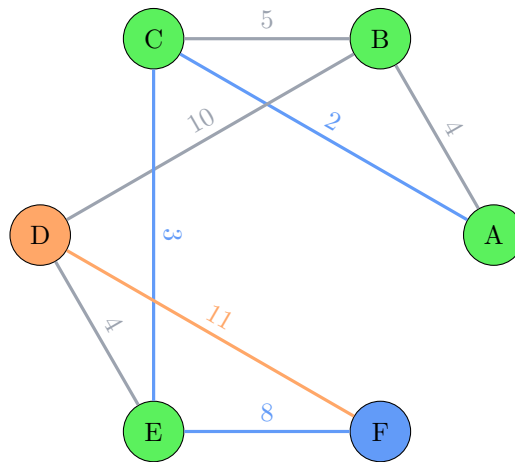


Dijkstra

Log

Looking at neighbor F | is going through D cheaper than what we already know?

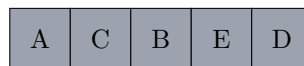
Graph



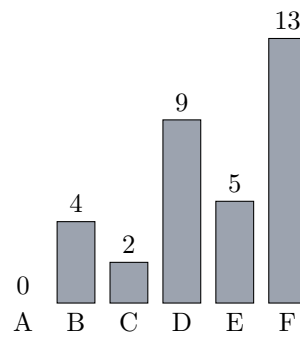
Open Set



Closed Set



Node Costs

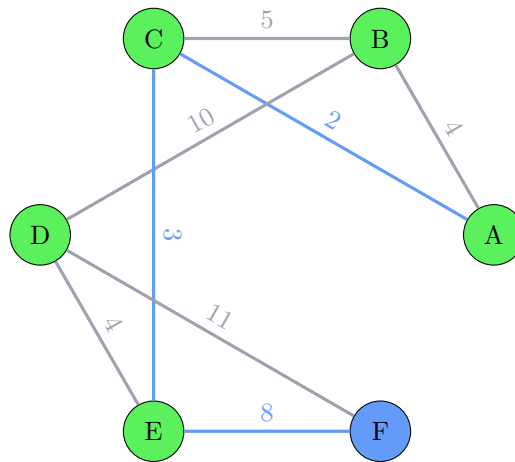


Dijkstra

Log

Sorting the open set so the lowest-cost node comes first.

Graph



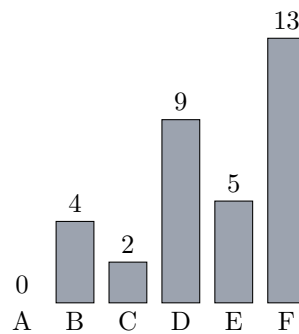
Open Set



Closed Set



Node Costs

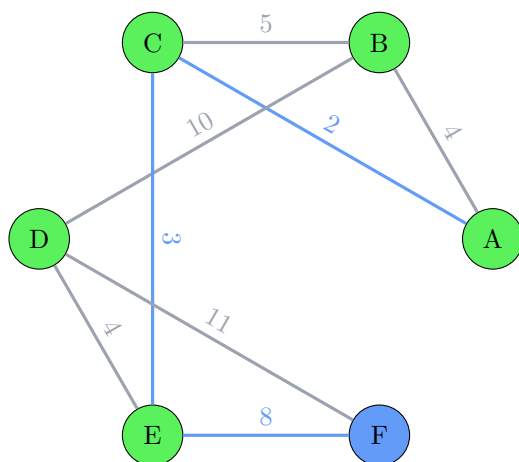


Dijkstra

Log

Grabbing the lowest-cost node from the open set.

Graph



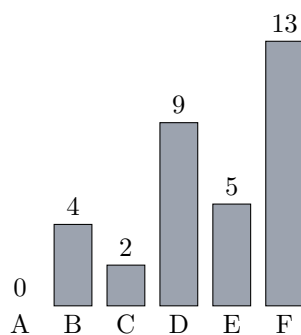
Open Set



Closed Set



Node Costs

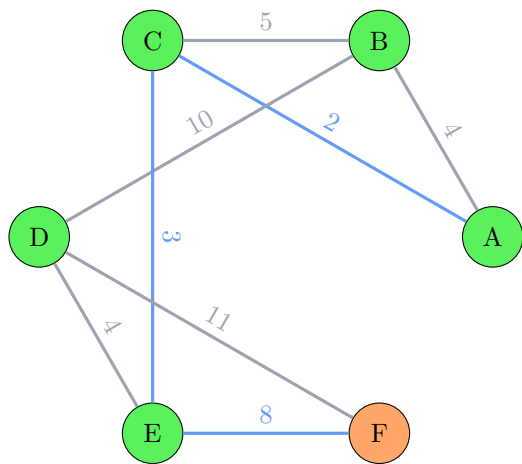


Dijkstra

Log

Visiting node F and moving it to the closed set.

Graph

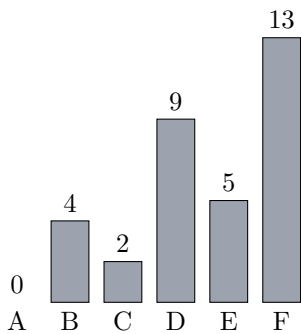


Open Set

Closed Set

A	C	B	E	D	F
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Node Costs

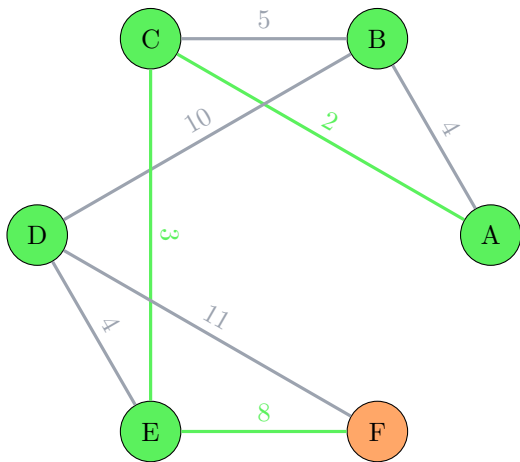


Dijkstra

Log

Reached the destination F | tracing back the shortest path.

Graph

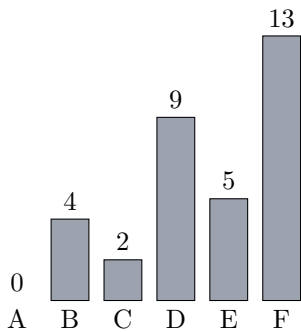


Open Set

Closed Set

A	C	B	E	D	F
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Node Costs

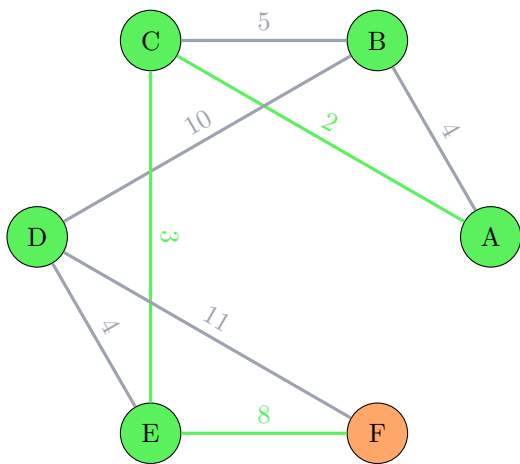


Dijkstra

Log

Dijkstra's algorithm has finished exploring the graph.

Graph



Open Set

Closed Set

A	C	B	E	D	F
---	---	---	---	---	---

Node Costs

